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Technical Note

Effects of cyclic loading on the mechanical properties of a granite

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1. Introduction

Many civil and mining related rock structures (e.g. tunnel walls, excavation roofs and ribs, bridge abutments, dam and road foundations) are subjected to dynamic loading due to, for example, drilling and blasting, muck loading and mechanical excavation. Although it is clear that the mechanical properties of rock under dynamic loads differ dramatically from those under static loads, the nature of dynamic failure in rock remains unclear, especially in cyclic loading condition [1].

Different materials show different responses when they are subjected to cyclic loading. Some materials become stronger and more ductile, while others become weaker and more brittle [2]. It is well known that rock strength decreases due the phenomenon of rock fatigue, and that the fatigue characteristics of rocks under dynamic cyclic loading play an important part in the prediction and control of earthquake and rock burst. This has resulted in significant research that focus on the dynamic mechanical features of rocks under different loading histories and loading conditions [3–8], and a comprehensive literature review on fatigue behavior in rocks has been reported by Bagde and Petros [2]. Since 2005, many researchers have studied the effect of cyclic loadings on the fatigue properties of rocks. Bagde and Petros [9], for example, performed a series of fatigue tests to investigate fatigue and dynamic energy variation in sandstone and conglomerate. The researchers concluded that development of fatigue failure in rocks was closely related to their petrographic, physical and mechanical properties, and that micro-fracturing was the main cause of fatigue failure.

Xiao et al. [7] suggested an inverted S-shaped nonlinear fatigue damage cumulative model based on the law of axial irreversible deformation development in rock. Xiao et al. [1] conducted cyclic loading tests on granite rock from the Heng Yang region in order to find a suitable damage variable to describe fatigue damage in rocks. They reported that the residual strain method was the most ideal method because of its distinct physical meaning, relevant description of the degradation behavior and recognition of the initial fatigue damage.

Fuenkajorn and Phueakphum [10] evaluated effect of cyclic loading on the mechanical properties of Maha Sarakham salt. They found that the salt compressive strength decreased by a power function as the

Table 1
Physico-mechanical properties of Alvand monzogranitic rock.

Parameters	UCS (MPa)	Tangent modulus (GPa)	Secant modulus (GPa)	Energy density (J/m^3)	Density (g/cm^3)	Porosity (%)
Maximum	112.98	29.95	23.12	0.27	2.74	1.40
Average	112.35	29.37	22.92	0.25	2.71	1.24
Minimum	111.63	28.63	22.58	0.23	2.69	1.06

Table 2
Summary of the fatigue tests characterestics.

Sample no.	Max. stress level	Min. stress level	Amplitude level	Frequency (Hz)	Number of failure cycle no.
A13	0.90	0.40	0.50	1	487
A41	0.90	0.00	0.90	1	520
A22	0.90	0.20	0.70	1	271
A47	0.95	0.25	0.70	1	104
A6	0.95	0.25	0.70	5	567
A23	0.95	0.25	0.70	0.1	46
A56	0.95	0.25	0.70	0.2	55
A20	0.85	0.17	0.68	1	2365
A17	0.87	0.17	0.70	1	1010

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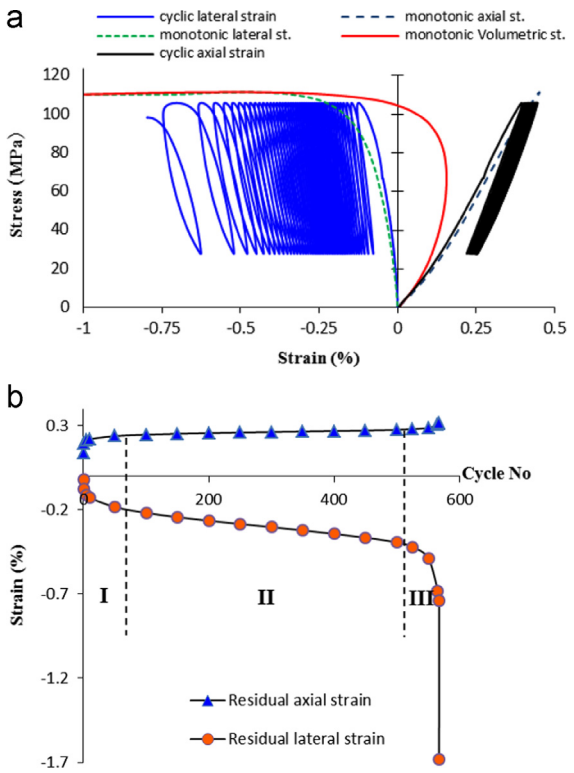


Fig. 1. (a) Stress–strain curves for monotonic (sample A2) and cyclic (sample A23) uniaxial compressive strength tests. (b) Three-step fatigue damage process in residual strain view (sample A6).

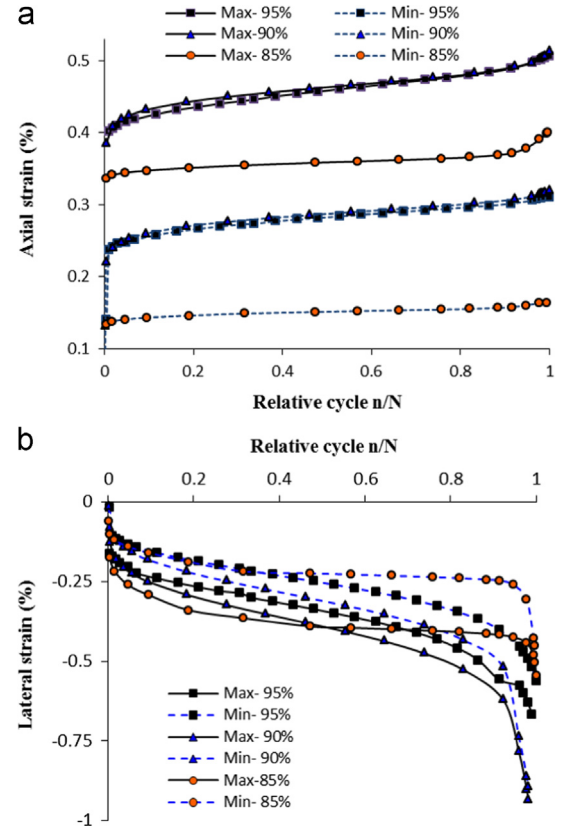


Fig. 2. Maximum and minimum (residual) strain damage processes in fatigue test using different stress levels: (a) axial strain versus relative cycle and (b) lateral strain versus relative cycle (frequency = 1 Hz).

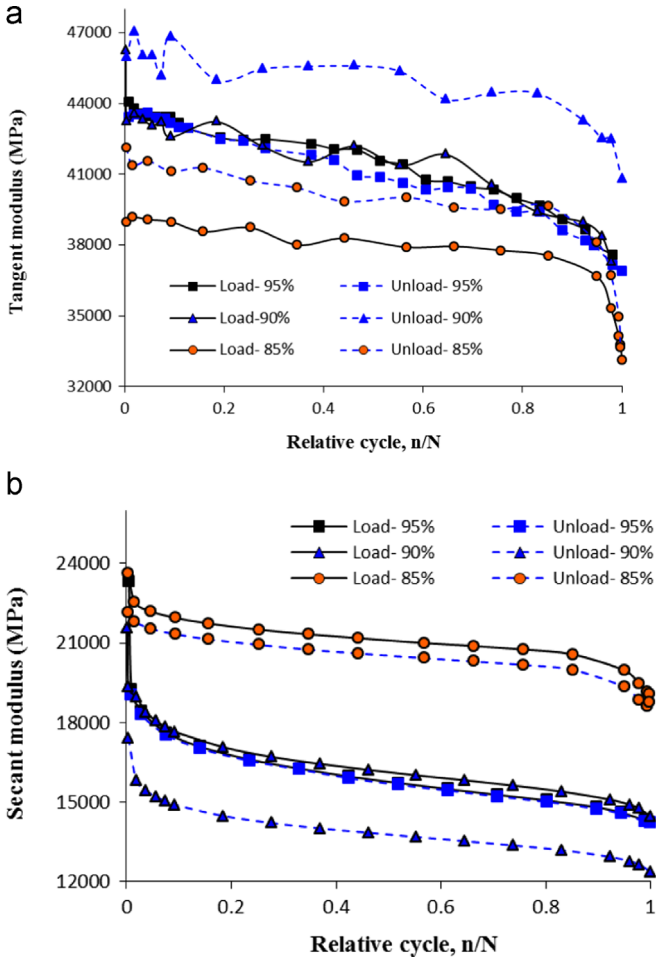


Fig. 3. Loading and unloading Young's modulus damage processes in fatigue test using different stress levels: (a) tangent modulus versus relative cycle and (b) secant modulus versus cycle (frequency = 1 Hz).

number of loading cycles increased, and that Young's modulus decreased slightly during the first few cycles then remained constant until failure. Liu and He [8] and Liu et al. [11] performed a series of laboratory tests to assess the effects of confining pressure on the mechanical properties and fatigue damage evolution of sandstone samples subjected to cyclic loading. Their results indicated that as confining pressure increased, the axial strain at failure increased, and concluded that the level of confining pressure had a significant influence on cyclic dynamic deformation and fatigue damage of the rock. However, available data on fatigue behavior remain insufficient for solving the practical tasks of predicting rock bursts and earthquakes. Obtained results are inconclusive and sometimes discordant. Granite is the most common intrusive igneous rock. Due to its high strength it has been widely used in the construction industry and deep mining operations are often conducted in the vicinity of granitic rocks. Therefore, it is crucial to understand granite's mechanical behavior, in particular, under cyclic loading conditions. The aim of the current work is to evaluate Alvand monzogranite rock fatigue behavior under different maximum loads, frequencies, and amplitudes to infer the fatigue damage process of the granitic rock.

2. Experimental set-up and rock properties

In order to carry out cyclic loading on granite, rock samples were chosen from the Alvand batholith outcrops, west of Iran. The outcrops are part of one of the largest plutonic bodies in the

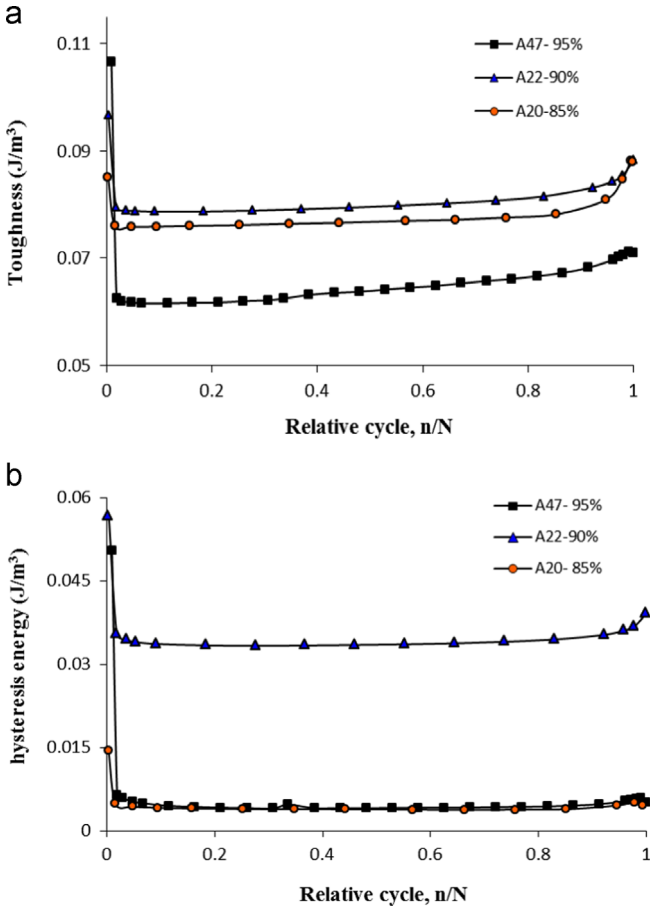


Fig. 4. Energy variation in fatigue processes using different maximum loads: (a) energy density versus relative cycle and (b) hysteresis energy versus related cycle (frequency = 1 Hz).

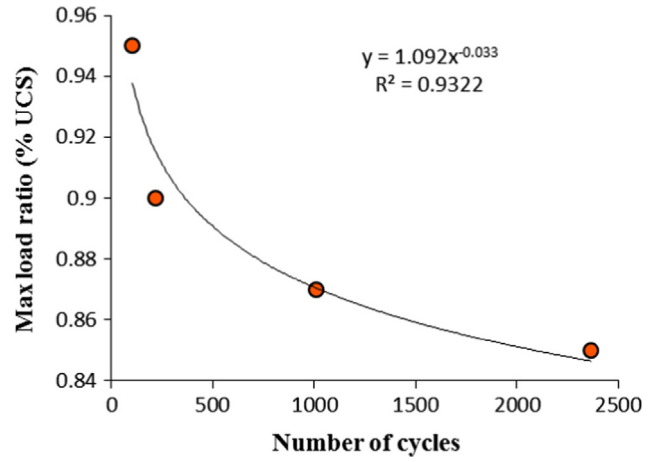


Fig. 5. Correlation between applied stress and number of cycles up to failure (amplitude = 0.7 and frequency = 1 Hz).

Sanandaj-Sirjan metamorphic belt. Alvand batholith consists of mafic (gabbro–diorite–tonalite) and intermediate (granite–granodiorite porphyroids) rocks, as well as hololeucocratic granitoids [12]. The bulk of the formation is porphyroid monzogranite.

Fatigue tests were conducted using an Instron 1342 servo hydraulic fatigue testing machine with a 300 kN capacity. The tests were performed under sinusoidal cyclic loading. The Instron controller consists of hardware components and software applications that provide closed-loop control of servo-hydraulic test equipment. The fatigue testing machine consists of a compression loading frame, an axial cyclic loading system and a data acquisition system. The data acquisition system consists of a signal conditioning, and an acquisition unit interfaced with a computer. Multiple or single data acquisition processes can collect data on all channels. The machine is equipped with an automated dynamic control mode for switching between any connected transducers. This fatigue testing machine is able to perform cyclic tests in both load and displacement control modes.

2.1. Testing methodology

Several core samples were prepared to perform this research. The core samples were prepared with a *L*/*D* ratio of 2.5 with an average diameter of 54 mm. The end faces of the core samples were prepared according to ISRM [13] standards. Before the fatigue tests, the physical and mechanical properties of the rocks were measured as shown in Table 1. The physico-mechanical index properties were determined according to the standard test procedures suggested by ISRM [13]. Uniaxial compressive strength

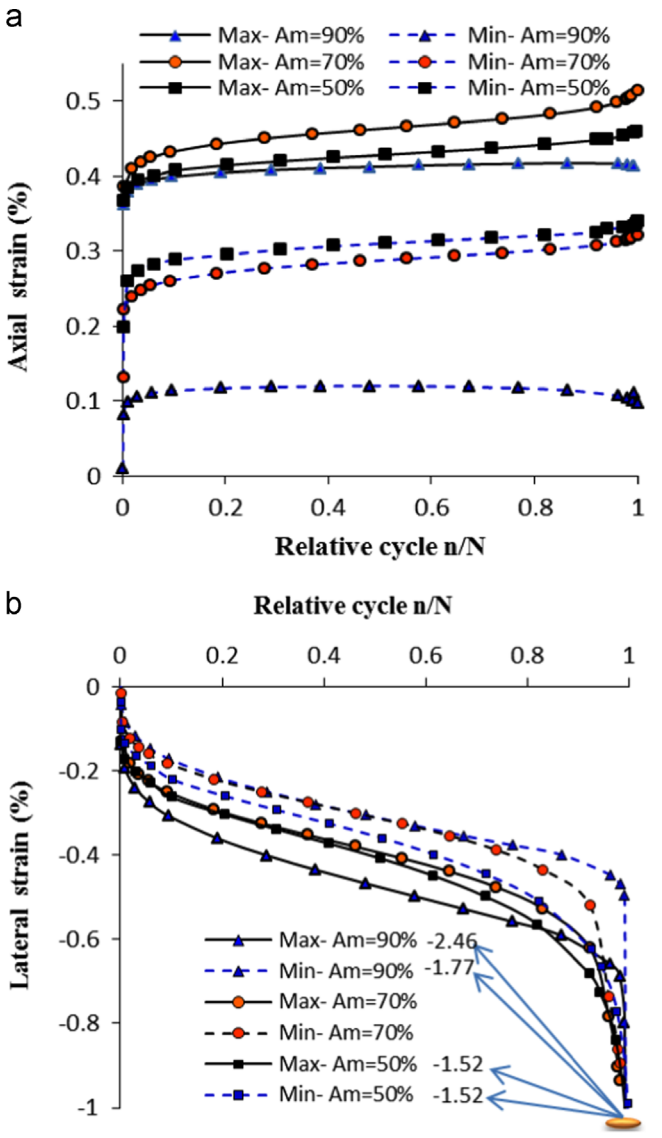


Fig. 6. Maximum and minimum (residual) strain damage processes in fatigue test using stress amplitude: (a) axial strain versus relative cycle and (b) lateral strain versus relative cycle (frequency = 1 Hz).

test (UCS) has been done on five core samples. The tests were performed in the load-control mode with a 1.6 kN/s loading rate. For each sample, four strain gauges have been used to record axial and lateral strains during the tests.

The tests were conducted to obtain the mechanical parameters of the rocks in static loading condition, and provided a reference for subsequent dynamic cyclic tests. All cyclic tests were performed in the stress control mode. Axial and lateral strains during the fatigue tests were recorded using strain gauges. Despite the fact that there was a change in material properties and crack growth, the load amplitudes were kept constant. The fatigue tests were conducted with four different maximum loads and at amplitudes with four frequency levels. The maximum stress level (the ratio of maximum cyclic stress to static strength) was varied, ranging from 0.95 to 0.85. The amplitude level (the ratio of amplitude stress to static strength) ranged from 0.50 to 0.70 and 0.90. Furthermore, in order to assess the effect of the loading frequency on fatigue behavior, four different frequencies 0.1, 0.2, 1 and 5 Hz were employed (Table 2). Because of some limitations in recording the 5 Hz data, the results of this test were not used in fatigue damage analysis and just used for assessment of loading frequency effect on fatigue life.

Initially, the specimen was loaded up to the mean stress level (σ_m) in the load-control mode with a 1.6 kN/s loading rate, as was used in the static tests. Then the fatigue test was performed with a given amplitude and frequency. It should be noted that the predetermined stress amplitude for the machine was equal to one-half of the total range ($A_m = \pm \sigma_a$). Finally, the results of the fatigue tests were analyzed according to strain spectrum, tangent and secant modulus, energy density and hysteresis energy.

3. Results and discussion

For a clear understanding of the effect of rock fatigue on damage mechanisms, a comparison between static and cyclic loading tests is shown in Fig. 1a. This figure highlights the fact that failure occurs below the maximum strength loading condition as a result of accumulative damage. Analysis of the fatigue test results showed that the fatigue failure consisted of three stages: fatigue crack formation (initiation phase I), stable crack propagation (uniform velocity phase II), and unstable crack propagation resulting in a sudden breakdown (accelerated phase III). These

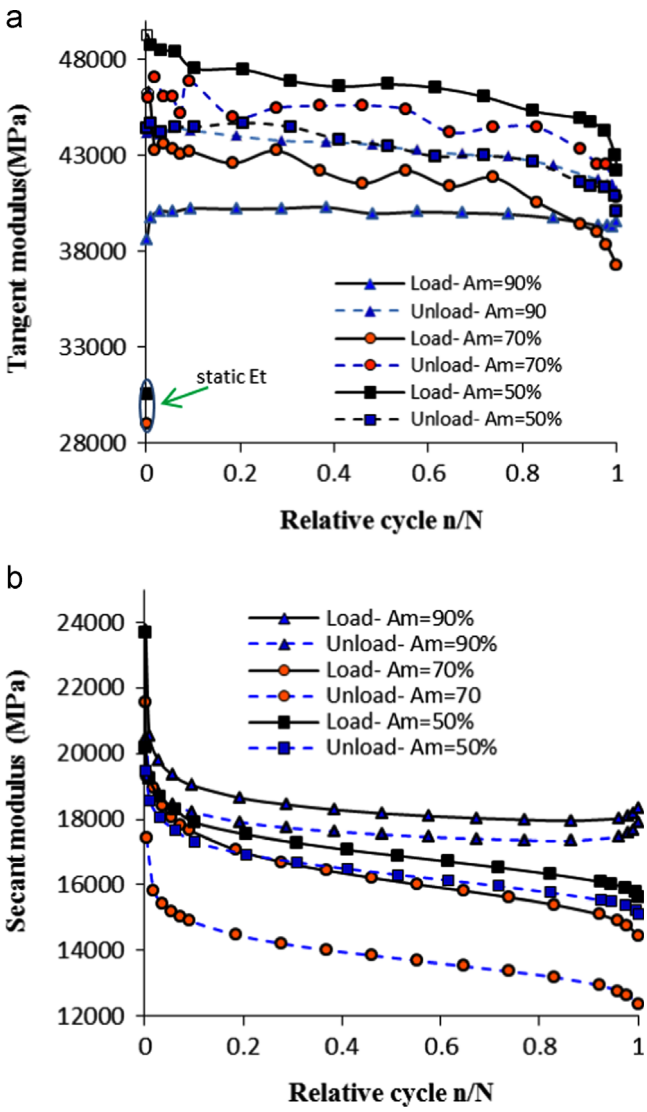


Fig. 7. Loading and unloading Young's modulus damage processes in fatigue test using different stress amplitudes: (a) tangent modulus versus relative cycle and (b) secant modulus versus cycle (frequency = 1 Hz).

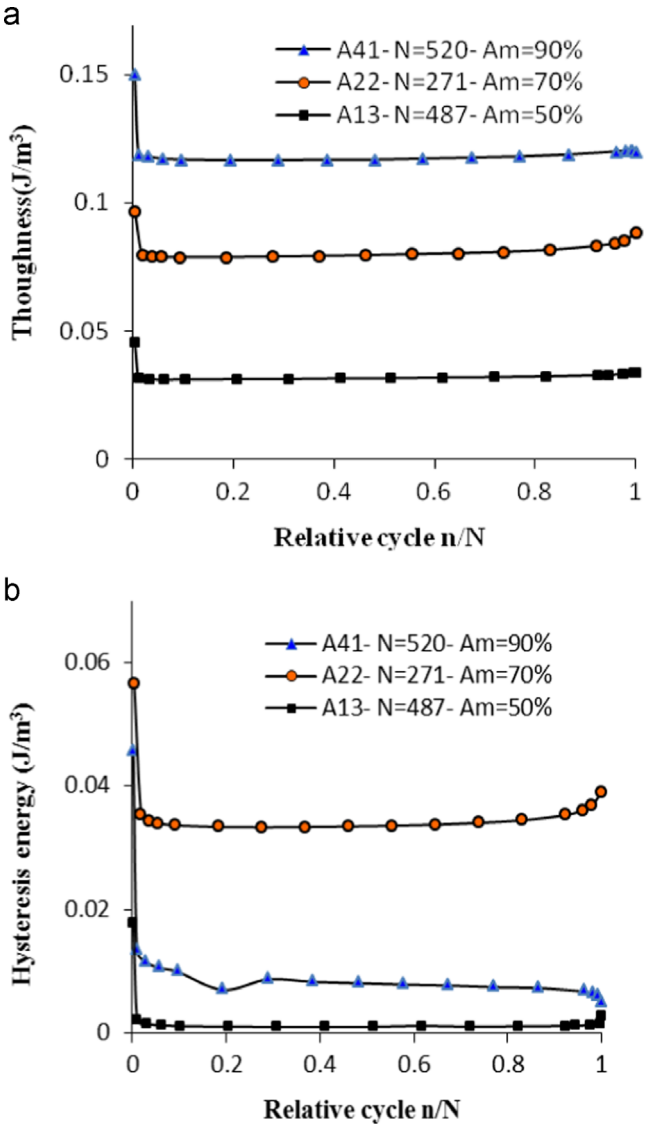


Fig. 8. Energy variation in fatigue processes using different stress amplitudes: (a) energy density versus relative cycle and (b) hysteresis energy versus related cycle (frequency = 1 Hz).

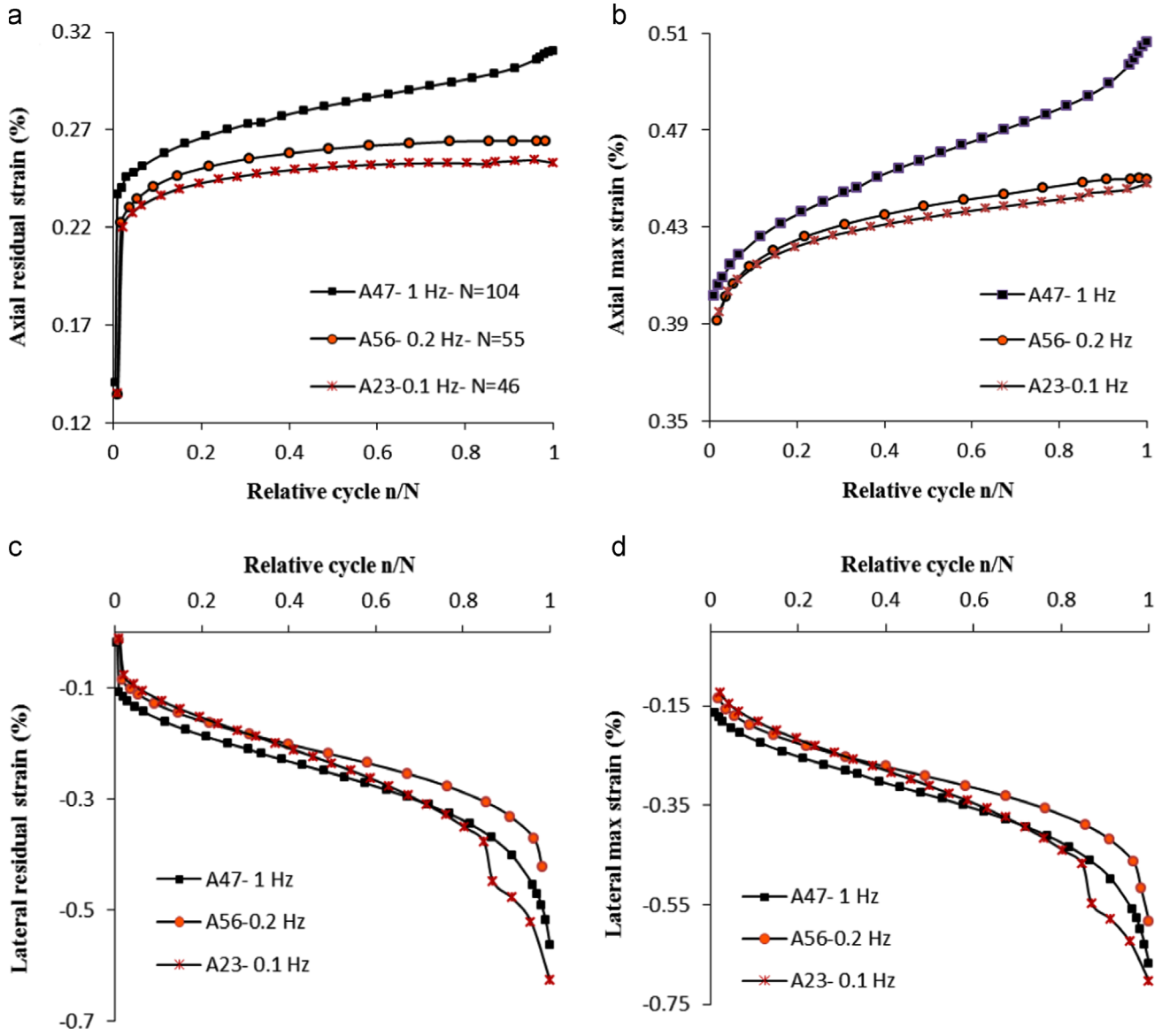


Fig. 9. Strain damage processes in fatigue test using same load condition and different frequencies: (a) axial residual strain versus relative cycle, (b) axial maximum strain versus relative cycle, (c) lateral residual strain versus relative cycle and (d) lateral maximum strain versus relative cycle.

three steps have also been reported by many researchers [1,7,8]. Fig. 1b shows the fatigue damage steps in both axial and lateral strain levels. The fatigue process is related to accumulative damage. Such damage leads to a progressive weakness of materials. In reality, the rock is not intact any longer after the first loading cycle because of weaknesses resulting from formation of microfractures.

Comparing the axial deformation, as shown in Figs. 1b, 4 and 5, it can be seen that lateral deformation is more sensitive to fatigue. Young's modulus, energy density, ultrasonic wave velocity, density and residual strength decrease with damage accumulation. On the other hand residual strain, electrical resistance and energy dissipation all increase with damage accumulation [1]. Variation of these parameters with time or cycle number can all be observed in the three stages of fatigue damage. Damage variables used in this study include a tangent modulus, secant modulus, hysteresis energy, energy density, maximum strain and residual strain. These variables are discussed in three different sections, including the effects of maximum stress, amplitude and frequency.

3.1. *The effect of maximum stress*

The relationship between the axial and lateral strain, fatigue damage and a relative cycle (ratio between cycle number and failure cycle number) at fixed amplitude of 0.70 and three different maximum stress levels of 0.95, 0.90, and 0.85 of uniaxial compressive strength is illustrated in Fig. 2. It should be noted that recording deformation properties of sample A17 with maximum load of 0.87 UCS was failed. The graphs indicate that at 0.95 loading stress levels, the second stage of fatigue life is dominant, whereas at 0.85 loading stress levels, the crack nucleation stages of fatigue life is dominant. Furthermore, the second stage slope of the accumulative damage from axial and lateral strain clearly decreases with decreasing stress levels.

Fig. 3 illustrates Young's modulus damage curves of the granitic rock. Both the tangent and secant modulus in this study were calculated at a stress level of 50% of the uniaxial strength. Fig. 3a uses a scatter pattern to show the descending trend of the tangent modulus under loading and unloading processes. The behavior

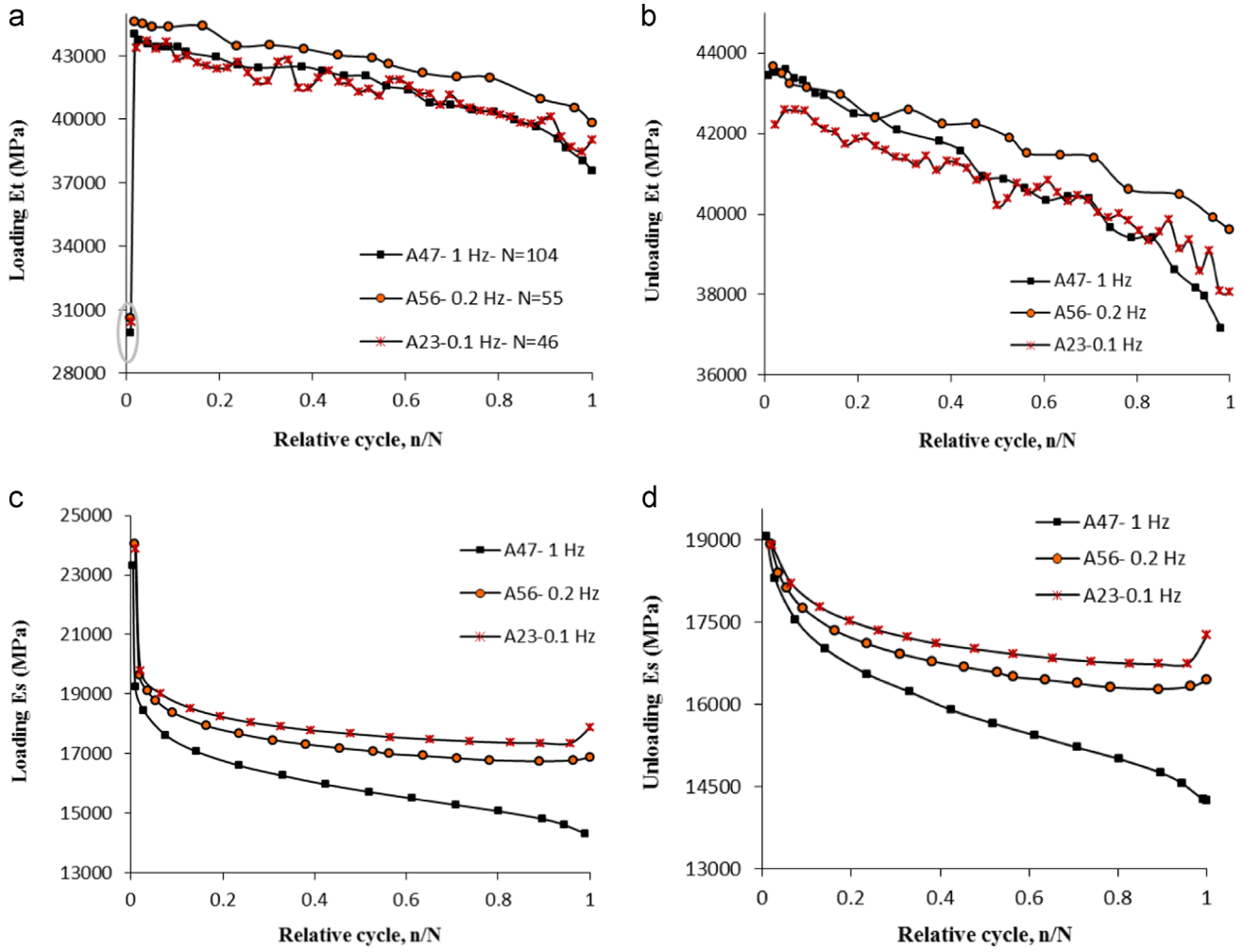


Fig. 10. Young's modulus damage processes in fatigue test using same load condition and different frequency: (a) loading tangent modulus versus relative cycle, (b) unloading tangent modulus versus relative cycle, (c) loading secant modulus versus cycle, and (d) unloading secant modulus versus cycle.

may be related to the calculation method and the loading condition, as well as the microstructure and behavior of the intact rock. In spite of the tangent modulus results, the three-stages of the damage process for the secant modulus in both the loading and unloading conditions is clear (Fig. 3b). The result is due to the method of calculation and the increase in axial strain with the increasing number of cycles. It should be noted that both the values and the slope of the second stage of damage for the secant modulus increase with increasing stress levels.

The energy density (areas under stress–strain curves ranging minimum and maximum stress) curves for the granitic rock at the three maximum stress levels are given in Fig. 4a. It should be noted that the energy density and hysteresis energy reported here have been calculated from maximum and minimum stress levels. There is a dramatic decrease of these parameters in the first few cycles due to the closure of pre-existing micro-fractures. In fact, during the initial cycle the rock behaves in a more ductile fashion than in the next few cycles. This behavior reflects the fact that granitic rocks are formed in very different thermal and pressure conditions than those common on earth's surface, resulting in the generation of micro-fractures due to unloading thermal expansion and pressure (relaxation fracture). Pre-existing micro-fractures resulting from this complex loading history close at the beginning of the testing cycles. Thereafter, energy density begins to increase slowly, then steadily, and finally rapidly. A similar behavior was found for hysteresis energy as well (Fig. 4b). Assessment of the effect of the maximum stress on fatigue life indicates that fatigue life increases in a power function with decreasing maximum stress (Fig. 5). The failure load obtained from the average static tests was reduced from 258 to 217 kN because of rock fatigue damage after 2364 cycles.

3.2. The effect of amplitude

The effect of stress amplitude on the damage process of the rock at a fixed maximum stress level of 0.90, with frequency 1 Hz and various amplitude ratios of 0.50, 0.7 and 0.90 are plotted in Figs. 6–8. The second stage of the whole fatigue process increases with increasing amplitude, while the average slope decreases (Fig. 6). This behavior may be a result of variations in the stresses in the plastic region. Since maximum stress is constant, increasing the amplitude leads to a decrease in minimum stress. This means that, time during which the stress acts in the plastic region decreases. Residual strain damage processes also indicate that initial damage decreases when stress amplitude is increased.

The tangent modulus shows a scatter pattern, especially at amplitude 70% of UCS. Similar to the maximum load effect, the descending trend of the loading and unloading tangent modulus is obvious. However, the three-stage fatigue process is not clear. Because the mean stress (σ_m) for amplitudes 0.50 and 0.70 is

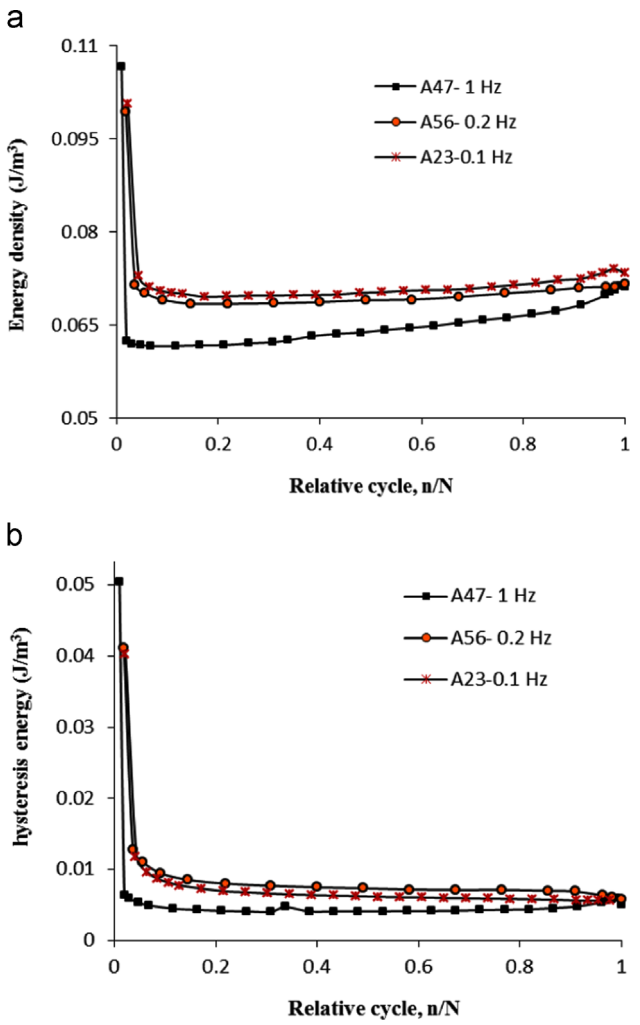


Fig. 11. Energy variation in fatigue processes using same load condition and different frequency: (a) energy density versus relative cycle and (b) hysteresis energy versus related cycle.

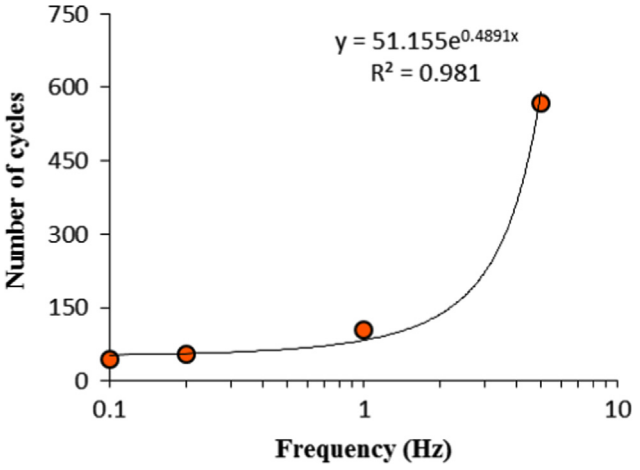


Fig. 12. Correlation between frequency and number of cycle to failure (maximum stress level of 0.95 and amplitude level of 0.7).

greater than 50% of UCS, the static tangent modulus is also included in order to compare it with the cyclic modulus (Fig. 7a). As is clear from Fig. 7a, the cyclic tangent modulus barely increases with respect to the static tangent modulus. The second damage stage also increases with the amplitude. Secant modulus is shown in Fig. 7b in which three-stage fatigue process can be distinguished. An anomaly is clear for the acceleration phase of the test with amplitude 0.90 that shows an inverse trend. This anomaly may have appeared because of local strain hardening processes. Generally, as amplitude increases, Young's modulus, in relation to accumulative damage, decreases. Regardless of decreases in the tangent modulus during the fatigue process, the tangent modulus of final cycle is greater than the static tangent modulus for this rock.

Fig. 8 shows energy density and hysteresis energy in fatigue damage with different amplitudes. Energy density increases with increasing amplitude because of decreases in minimum stress. Generally, for the initial cycles, a dramatic decrease is very clear. As noted earlier, this is because of the closing of the pre-existent micro-fractures and slow rate of loading up to mean stress (σ_m). Generally, the same results were achieved for the accumulated damage caused by hysteresis energy.

3.3. The effect of loading frequency

In order to understand the effect of loading frequency on the fatigue process in cyclic loading, four tests were performed using loading frequencies of 0.1, 0.2, 1 and 5 Hz. It should be noted that the tests are conducted under only one stress level (0.95) and one amplitude (0.7) level. Strain variation and its dependence on the loading frequency are illustrated in Fig. 9 that shows a general increasing trend for axial maximum and residual strain as frequency increases (Fig. 9a and b). Time factor plays a significant and important role here, as the load has more time to act at low frequencies than at higher frequencies. This relates to the phenomena called the dynamic loading effect, or rate-dependent behavior, and is well known in rock mechanics. It is well documented that high frequency will cause the increase in fatigue life and the decrease in the fatigue cracks propagation rate. It is interesting to note that the acceleration phase (III) in both axial and lateral strain accumulated damages is highlighted by increasing frequency. At low frequencies (0.1 and 0.2 Hz) the rock shows brittle behavior and the acceleration phase is not clear. It appears that with decreasing frequency, crack growth has enough time to coalesce and creates a failure surface at the maximum load.

Tangent and secant modulus variations for different frequencies, in both loading and unloading are illustrated in Fig. 10. The tangent modulus shows a scattering pattern and descending trend in both loading and unloading conditions. It should be noted that, three-stage damage process are not clear at different frequencies. As can be seen from Fig. 10, with increasing frequency, the emergence of an acceleration phase for both tangent and secant modulus is highlighted. This trend is a result of the fact that at high frequencies maximum stress is not being applied for long enough to create a failure surface, and therefore a few more cycles are needed to break the rock.

Fig. 11 shows that energy density and hysteresis energy vary during the fatigue process with varying frequencies. Generally at different frequencies, maximum loads and amplitudes, the levels of energy density and hysteresis energy in the initial cycles decreased. Energy density and hysteresis energy decreased as frequency increased. Furthermore, the results showed a general trend of decreasing slope of second stage of fatigue damage with decreases in frequency. The results show an exponential increase in fatigue life at higher loading frequencies (Fig. 12). The reason for the increase in fatigue life with loading frequency could be that the load increases very rapidly that there is no time for flaws or cracks to nucleate or propagate. Frequency will cause the increase in fatigue life and the decrease in the fatigue cracks propagation rate.

4. Conclusions

Alvand monzogranitic rocks were subjected to uniaxial cyclic loading under different maximum loads, amplitudes and frequencies in order to understand their fatigue behavior. The following conclusions were drawn from this research.

Accumulated fatigue damage occurs in an obvious three-stage process. This is the result of the micro-fracturing mechanism in the fatigue process. By comparing axial and lateral strain damages, it was found that crack propagation occurred in the loading direction and crack opening occurred in the lateral direction.

Analysis of fatigue damage at different load levels shows that when the maximum load level is decreased, crack nucleation occurs during a large proportion of the whole fatigue life, while crack growth occurs at higher stresses.

Fatigue life increases in a power function with decreasing maximum stress. This is due to a decrease of plastic deformation. In addition to increases in fatigue life, the steady stage slope decreases.

Tangent modulus in different loading conditions shows a decreasing trend with a scattering pattern. The secant modulus, on the other hand, shows three stages of accumulative damage during the fatigue process.

Energy density and hysteresis energy variation in different loading conditions show a decrease in the first few cycles and start to increase up to failure. In fact these two parameters have a short decreasing trend phase, dominated by uniform increasing phase and short rapid increasing phase.

Fatigue life increases in an exponential function with increasing loading frequency. This is due to decreasing time for crack growing.

Because of the brittle behavior of the rock and the low rate of loading at low frequencies (0.1 and 0.2) the acceleration phase in the accumulative damage process could not be traced.

The residual axial and lateral strain of the rock decreased with increasing amplitude. This trend was also found for second stage slopes of strain and energy density accumulative damage.

After a few initial cycles, the rocks exhibited almost elastic behavior. However, with the increasing numbers of cycles elasto-plastic behavior is observed. It can be inferred that with increasing cycle numbers, the yield stress level decreased and plastic behavior becomes dominant in each cycle.

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